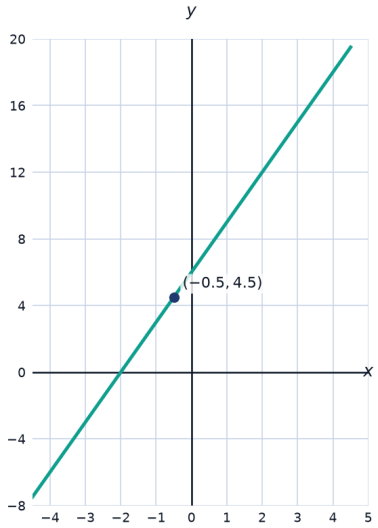
Mensuration

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

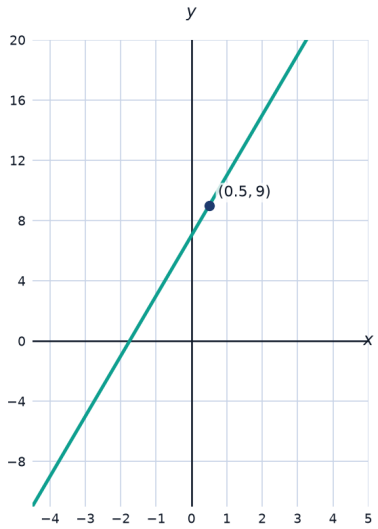
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in percentage of quantities, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in percentages with technology, evaluate $f(2)$.



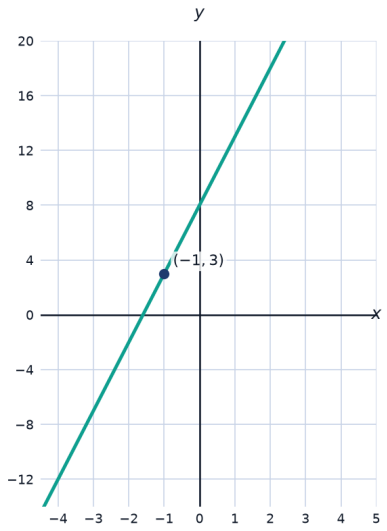
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for percentages with technology consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in finding the whole (extended), evaluate $f(2)$.

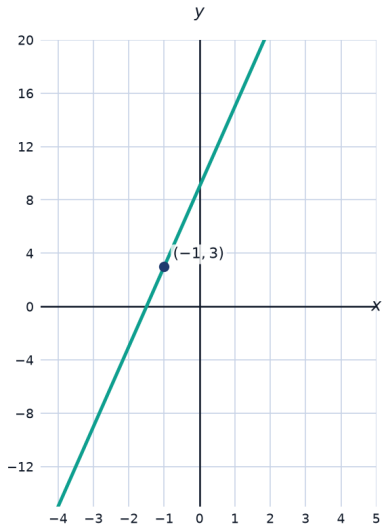


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for finding the whole (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 6x + 9$ in percentage increase and decrease, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for percentage increase and decrease consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

£800 is invested at 3.5% compound interest per year. Find the amount after 4 years, to the nearest penny.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the magnitude of $\mathbf{v} = \begin{pmatrix} -5 \\ 12 \end{pmatrix}$ and hence find a unit vector in the direction of $\mathbf{v}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

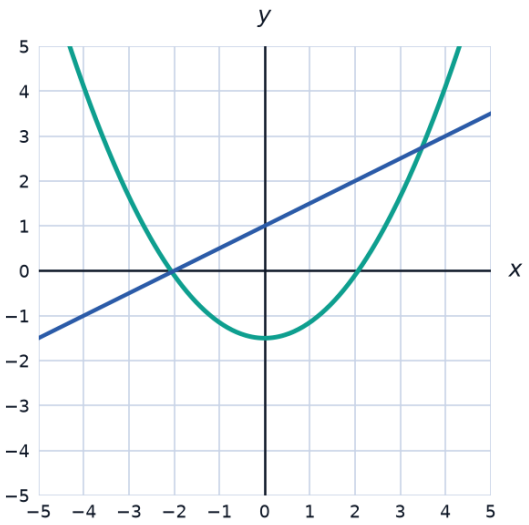
A tower PQ stands vertically. From point A on level ground, 50 m from the base Q, the angle of elevation of P is 32°. Find the height of the tower.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The diagram shows a sector of a circle with radius r. The angle at the centre is θ radians. If the arc length is USD 6$cmandtheradiusisUSD3$ cm, find the angle θ in radians.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle of radius $10\ \text{cm}$, chord AB subtends an angle of $\theta = \frac{2\pi}{3}$ radians at the centre O. Find: (a) the exact length of chord AB, (b) the exact area of the minor segment bounded by chord AB and minor arc AB, in terms of π and $\sqrt{3}$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number**

A stone is thrown vertically upwards with speed $15\ \text{m/s}$ from the top of a cliff $40\ \text{m}$ above sea level. Taking $g = 9.8\ \text{m/s}^2$ and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Sets & Venn Diagrams**

For what value of k will the system have no solution?

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Algebra & Graphs**

A small particle of weight $24\ \text{N}$ is held in equilibrium by two light inextensible strings. String A makes an angle of 30° with the upward vertical, and string B makes an angle of 60° with the upward vertical. Find the tensions T_A and T_B in the two strings.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry**

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry**

A cone and a cylinder have identical base radii of $6\ \text{cm}$. If both solids have the same volume of $180\pi\ \text{cm}^3$, find the height of the cylinder and the height of the cone.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry**

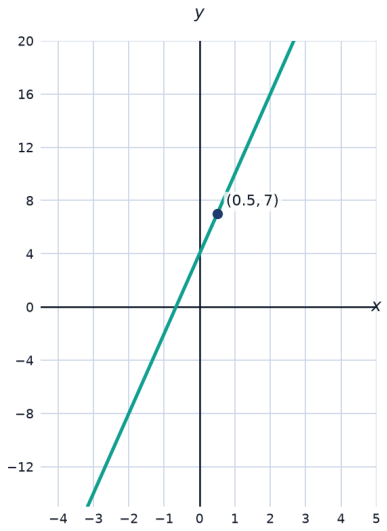
The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of C. (b) Find the equation of the tangent to C at the point $P(6, 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in converting between percentages, fractions, and decimals, evaluate $f(2)$.



a. 16

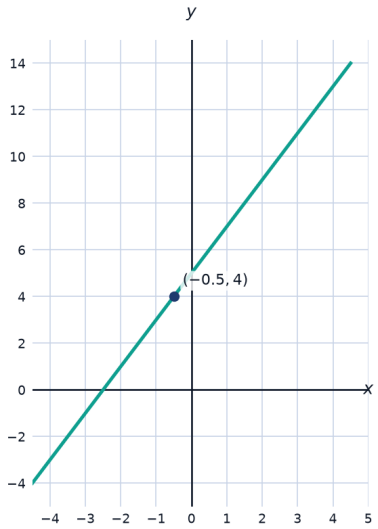
b. 22

c. 12

d. 20

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in quantities as percentages, evaluate $f(2)$.



a. 9

b. 11

c. 4

d. 13

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for converting between percentages, fractions, and decimals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for quantities as percentages consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

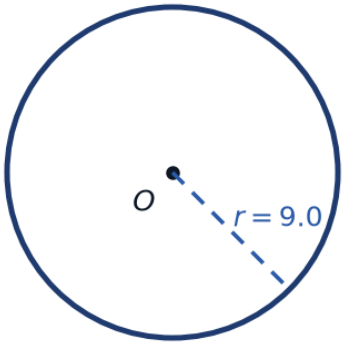
b. 18.5

c. 19.5

d. 21.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An arc of a circle of radius 9 cm subtends an angle of 120° at the centre. Find the arc length in terms of π.



a. 6π cm

b. 3π cm

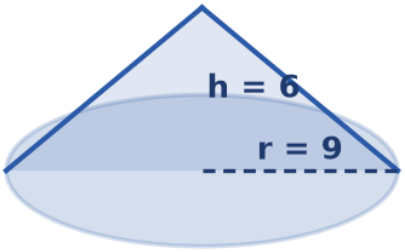
c. 9π cm

d. 12π cm

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A cone has radius 9 cm and height 6 cm. Find its volume in terms of π.



a. 162π cm^3

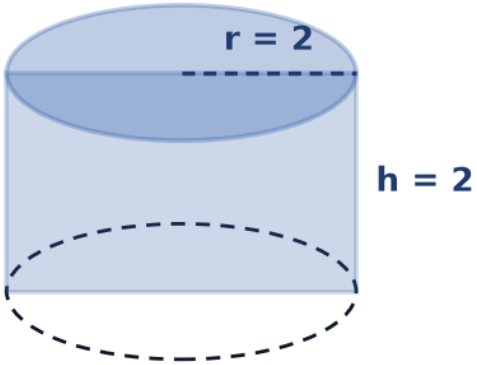
b. 152π cm^3

c. 174π cm^3

d. 159π cm^3

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A closed cylinder has radius 2 cm and height 2 cm. Find its total surface area in terms of π.



a. 16π cm^2

b. 25π cm^2

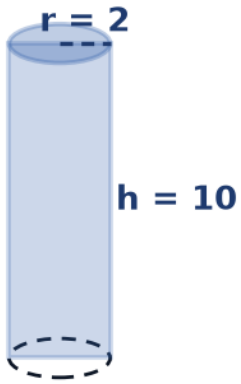
c. 27π cm^2

d. 28π cm^2

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A cylinder has radius 2 cm and height 10 cm. Find its volume in terms of π.



a. 50π cm^3

b. 29π cm^3

c. 40π cm^3

d. 46π cm^3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A cuboid has a length of 8 cm, a width of 5 cm, and a height of 3 cm. Calculate the volume of the cuboid.

a. 120 cm^3

b. 68 cm^3

c. 40 cm^3

d. 128 cm^3

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A frustum of a right cone has top radius 3 cm, bottom radius 6 cm, and vertical height 4 cm. What is the volume of this frustum in terms of π?

a. 84π cm^3

b. 76π cm^3

c. 96π cm^3

d. 108π cm^3

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a circle of radius 12 cm, a chord subtends a central angle of 90°. Calculate the area of the minor segment. Give your answer to 1 decimal place.

a. 41.1 cm²

b. 36.0 cm²

c. 45.2 cm²

d. 52.8 cm²

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Calculate the total surface area of a solid hemisphere with radius 5 cm. Give your answer in terms of π.

a. 50π cm^2

b. 75π cm^2

c. 100π cm^2

d. 25π cm^2

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Calculate the total surface area of a cone with radius 7 cm and slant height 25 cm. Give your answer in terms of π.

a. 175π cm^2

b. 224π cm^2

c. 245π cm^2

d. 300π cm^2

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A copper wire of diameter 2 mm has a total volume of 50π cm^3. What is the length of the wire in metres?

a. 50 m

b. 5 m

c. 500 m

d. 25 m

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A right pyramid has a square base of side 6 cm and a vertical height of 10 cm. What is its volume?

a. 120 cm³

b. 360 cm³

c. 60 cm³

d. 180 cm³

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has radius 6 cm. Calculate the area of a sector with central angle 60°. Give your answer in terms of π.

a. 6π cm^2

b. 12π cm^2

c. 36π cm^2

d. 2π cm^2

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A solid hemisphere has radius 4 cm. What is its TOTAL surface area (including the flat circular base) in terms of π?

a. 48π cm^2

b. 32π cm^2

c. 64π cm^2

d. 16π cm^2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry

A sector of a circle has central angle 40° and area 16π cm^2. Find the radius of the circle.

a. 12 cm

b. 6 cm

c. 8 cm

d. 16 cm

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry

A sphere has radius 3 cm. Calculate its volume in terms of π.

a. 12π cm^3

b. 27π cm^3

c. 36π cm^3

d. 108π cm^3

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Trigonometry

A cone has radius 3 cm and vertical height 4 cm. What is its curved surface area in terms of π?

a. 15π cm^2

b. 12π cm^2

c. 30π cm^2

d. 36π cm^2

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Transformations & Vectors

Two similar solid cylinders have heights in the ratio 2 : 3. If the total surface area of the smaller cylinder is 40 cm^2, what is the total surface area of the larger cylinder?

a. 90 cm^2

b. 60 cm^2

c. 80 cm^2

d. 135 cm^2

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Probability & Statistics

Calculate the total perimeter of a sector with radius 7 cm and angle 60° (to 1 decimal place).

a. 21.3 cm

b. 26.3 cm

c. 7.3 cm

d. 14 cm

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Functions

A solid cone has radius 5 cm and slant height 13 cm. Calculate its vertical height.

a. 10 cm

b. 11 cm

c. 12 cm

d. 14 cm

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number

The curved surface area of a cylinder with radius 5 cm is 70π cm^2. What is the height of the cylinder?

a. 7 cm

b. 14 cm

c. 5 cm

d. 10 cm

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A rectangular swimming pool is 25 m long, 10 m wide, and has a uniform depth of 2 m. How many cubic metres of water does it hold when completely full?

a. 500 m^3

b. 250 m^3

c. 1000 m^3

d. 750 m^3

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